



AMDNet23: Hybrid CNN-LSTM deep learning approach with enhanced preprocessing for age-related macular degeneration (AMD) detection

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ABSTRACT

In the context of a growing global population, an automated framework for disease detection can greatly assist medical professionals in diagnosing ocular diseases. This framework offers advantages such as accurate, stable, and rapid outcomes, thereby enhancing the success rate of early disease detection. The primary objective of this study was to enhance the quality of fundus images using an adaptive contrast enhancement algorithm (CLAHE) and Gamma correction as preprocessing techniques. CLAHE is employed to heighten the local contrast of fundus images, while Gamma correction enhances the intensity of relevant features. This research adopts a deep learning approach that integrates convolutional neural networks (CNNs) and both short-term and long-term memory (LSTM) mechanisms. The purpose of this combination is to automatically detect aged macular degeneration (AMD) in fundus ophthalmology. In this mechanism, CNNs are utilized to extract pertinent features from the images, and LSTM is subsequently employed to discern these extracted features. To validate the effectiveness of the proposed method, a diverse dataset of 2000 experimental fundus images is collected from various sources. These images are categorized into four distinct classes in an equitable manner. Quality assessment techniques are then applied to this dataset. The hybrid deep AMDNet23 model proposed in this study achieves successful detection of AMD ocular disease with an achieved accuracy of 96.50 %. Moreover, the system's prowess is benchmarked against 13 other pre-trained CNN models, effectively illuminating its supremacy in AMD ocular disease diagnosis within the realm of fundus imagery datasets. This exhaustive comparison underscores the method's immense potential and reaffirms its position at the forefront of cutting-edge ocular health diagnostics.

1. Introduction

Over the past two decades, ocular diseases (ODs) that can cause blindness have become extremely widespread. ODs encompass a wide range of conditions that can affect various components of the eye, including the corneal tissue, lens, retina, optic nerves, and periorbital tissues. Ocular diseases include abnormalities such as cataracts, untreated nearsightedness, trachoma, age-related macular degeneration, and diabetes-associated retinopathy. These conditions play a substantial role in global retinal degeneration and visual impairment (Ackland et al., 2017). The worldwide prevalence of nearsightedness or farsightedness vision deficiency affects over 2.2 billion individuals (Vision Impairment & Blindness, 2022). Approximately half of the total cases, amounting to at least 1 billion people, as reported by the World Health Organization (WHO), suffer from vision impairments that could have

been prevented or remain unattended. Among these individuals, around 88.4 million have untreated refractive errors leading to mild to extensive distance impaired vision, nearly ninety-four million have cataracts, and eight million individuals are affected by age-related macular degeneration, diabetic retinopathy (3.9 million) (Vision Atlas, 2023). Despite significant investment, the number of individuals living with vision loss might increase to 1.7 billion by 2050, up from the 1.1 billion people recorded in the year 2020. Age-associated macular degeneration (AMD) predominantly affects the older demographic, resulting in the gradual deterioration of the macula, a crucial part of the retina responsible for central vision. The consequences of AMD manifest as central vision abnormalities, including blurred or distorted vision, which significantly impede various daily activities (Common Eye Disorders & Diseases, 2022). AMD constitutes a significant global health challenge, affecting an estimated 196 million individuals, as reported by the World Health

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Organization. This ocular ailment, which poses a substantial threat to vision, is dichotomized into two primary categories: Early AMD, encompassing a predominant 90 % of cases, and Late AMD, representing the remaining 10%. Timely identification assumes a critical role in the effective management of AMD, with expeditious intervention correlating with an impressive 90 % success rate in recovery (Wong et al., 2014).

Accurate and early identification of AMD disease plays a pivotal role in preventing irreversible damage to vision, facilitating timely treatment, and preserving ocular health. The advancement of machine learning techniques (Lilhore et al., 2022; Rana et al., 2022) has reached a stage where automated systems offer significant advantages in the early detection of aged macular degeneration compared to manual methods (Sarki et al., 2020). In the realm of diagnosing eye diseases, digital eye images and computational intelligence (CI)-based technologies have emerged as invaluable tools, empowering doctors to enhance their diagnostic capabilities (Kumar & Gunasundari, 2023). Various approaches are employed in medical imaging (Caroprese et al., 2022), including fundus photography (Iqbal et al., 2022), optical coherence tomography (OCT), and imaging modalities specifically designed for ocular examinations. These imaging technologies enable detailed visualization and analysis of ocular structures, thereby facilitating the identification of characteristic features and abnormalities associated with age-related macular degeneration [AMD] diseases.

Researchers have undertaken a crucial task within ophthalmology, aiming to enhance the early detection and diagnosis of age-related macular degeneration (AMD) ocular disease through the utilization of fundus images. The research community has directed its focus towards the application of deep learning techniques (Tan et al., 2018; Sogawa et al., 2020; Serener & Serte, 2019). This encompasses the convergence of vision and computing (Yim et al., 2020; Schmidt-Erfurth et al., 2018), alongside the implementation of predictive machine learning methods (Schmidt-Erfurth et al., 2018; Bogunović et al., 2017). These endeavors have led to the development of robust classification models, effectively categorizing retinal images into distinct AMD disease categories with a high degree of accuracy. The integration of deep learning (Muni Mishra et al., 2022) methodologies assumes a pivotal role in the precise classification of diverse ocular diseases. This, in turn, contributes significantly to the progression of intelligent healthcare practices within the realm of ophthalmology (Jaiswal et al., 2021).

This paper endeavors to present a pioneering system founded upon a deep learning framework, wherein the synergistic capabilities of CNN and LSTM networks are harnessed for the automated detection of AMD through fundus photography. In this innovative approach, CNN fulfills the pivotal role of extracting salient fundus features, while LSTM assumes the critical responsibility of classifying AMD based on the extracted features. The intrinsic memory within the LSTM network empowers it to assimilate knowledge gleaned from significant experiences over extended periods, contributing to its robust performance. In the architecture of fully interconnected networks, each layer is intricately linked, with nodes within layers remaining unconnected. In contrast, the LSTM nodes establish connections within a directed graph, following a temporal sequence that serves as a specialized input form Wang et al. (2020). This dynamic amalgamation of the two-dimensional CNN and LSTM systems profoundly enhances the classification of AMD within the field of ophthalmology, thereby facilitating informed clinical decision-making. The dataset utilized in this study is meticulously curated from diverse sources, and the preprocessing techniques are thoughtfully applied to enhance image quality, thereby optimizing the efficiency of AMD classification. Moreover, in a comprehensive evaluation, our approach's efficacy is benchmarked against a selection of 13 pre-trained deep learning models, including ViT-B16, MobileViT-XXS, InceptionResNetV2, EfficientNet-B7, EfficientNet-B6, DenseNet121, DenseNet169, DenseNet201, InceptionV3, MobileNetV2, VGG16, VGG19, and ResNet50. This thorough comparison underscores the supremacy of our proposed framework and underscores its potential for

accurate AMD diagnosis.

The main contributions of this study are summarized as:

- i. **Integrated Deep Learning Framework:** Integration of CNN and LSTM networks enables automated AMD identification from fundus images, enhancing diagnostic accuracy.
- ii. **Enhanced Diagnostic Precision:** The combined model significantly improves AMD classification, aiding clinicians in making well-informed decisions for better patient care.
- iii. **Benchmarking Against Pretrained Models:** Rigorous comparison with 13 pretrained models validates the framework's superiority, showcasing its potential for cutting-edge AMD diagnosis.
- iv. **Quality Enhancement and Dataset Diversity:** This research introduces preprocessing techniques involving Adaptive Contrast Enhancement Algorithm (CLAHE) and Gamma correction to elevate the quality of fundus images. Additionally, the diverse dataset collected from various sources ensures the robustness and generalizability of the proposed framework, further enhancing its effectiveness in real-world clinical settings.

The remainder of this article is organized as follows: Section II delves into the related works pertinent to this research. Section III articulates the proposed methodology, encompassing data collection, preprocessing techniques, and a comparison with existing models. In Section IV, we present the experimental findings and discussions, which include state-of-the-art and transfer learning comparisons. followed by Section V, which addresses Study Limitations and Scope for Future Research. Finally, the conclusion is presented in Section VI.

2. Literature review

In the pursuit of identifying ocular diseases, researchers have harnessed the power of deep learning techniques. These methods leverage fundus ophthalmology to facilitate the diagnosis of ocular diseases. This reviewed literature presents cutting-edge systems that have developed deep learning techniques for detecting AMD, diabetes, and cataracts.

Sahoo et al. (2023) introduced an innovative ensemble-based prediction model named weighted majority voting (WMV) for the exclusive diagnosis of Dry-AMD. This approach intelligently amalgamates predictions from diverse base classifiers, assigning specific weights to each classifier. Muthukannan (2022) presented a computer-aided approach that leverages the Flower Pollination Optimization Approach (FPOA) in conjunction with a CNN mechanism for preprocessing, specifically utilizing the maximum entropy transformation on the ODIR public dataset. The model's performance was benchmarked against other optimized models, exhibiting superior accuracy at 95.27 %. In a study by Serener and Serte (2019), the objective was to employ OCT images and deep neural networks for the detection of both dry and wet AMD. The outcomes unveiled that the eighteen-layer ResNet model accurately identified AMD with an astounding accuracy of 94 %, whereas the AlexNet model achieved an accuracy of 63 %.

Various deep learning methods have been explored for cataract detection; however, limitations in feature extraction and preprocessing often hinder their efficacy. Kumar and Gupta (2023) proposed multiple models to enhance clinical decision-making among ophthalmologists. Paradisa et al. (2022) applied the Concatenate model to fundus images, utilizing Inception-ResNet V2 and DenseNet121 for feature extraction, followed by MLP for classification, achieving an average accuracy of 91 %. Faizal et al. (2023) developed an automated cataract detection algorithm using CNN, achieving a notable accuracy of 95 % through the analysis of visible wavelength and anterior segment images. Pahuja et al. (2022) enhanced model performance through data augmentation and feature extraction methods. They employed CNN and SVM models to detect cataracts in a dataset comprising normal and cataract retinal images, attaining high accuracy rates of 87.5 % for SVM and 85.42 % for CNN. Chaudhary and Kumar (2022) utilized a CNN model for

diagnosing cataract pathology using digital camera images. Mahmud et al. [Khan et al. \(2021\)](#) employed color fundus images for cataract detection, illustrating the diverse strategies and approaches employed in the quest for accurate and efficient cataract identification.

A diverse array of computer vision engineering techniques ([Aggarwal et al., 2023](#)) are employed for the automated prediction of Diabetic Retinopathy (DR) occurrences and stages. [Mondal et al. \(2023\)](#) developed a collaborative deep neural system for automated diabetic retinopathy (DR) diagnosis, using modified DenseNet101 and ResNeXt models. Results on APTOS19 and DIARETDB1 datasets show elevated accuracy of 86.08 % for five classes and 96.98 % for two classes. Additionally, an ML-FEC model with pretrained CNNs (ResNet50, ResNet152, SqueezeNet1) achieves DR detection accuracy of up to 94.40 %, demonstrating its clinical potential and suitability for large-scale screening. Notably, [Babenko et al. \(2022\)](#) leverage a novel CNN model to achieve multi-class categorization of retinal fundus images from a publicly accessible dataset, attaining an accuracy of 81.33 % for diabetic eye disease. [Saranya et al. \(2023\)](#) extended the UNet architecture, achieving impressive accuracy in red lesion identification (95.65 %) and DR severity classification (94 %). The approach is validated with public datasets-IDRiD (99 % specificity, 89 % sensitivity) and MESSIDOR (94 % accuracy, 93.8 % specificity, 92.3 % sensitivity)-highlighting its robustness and practical viability.

3. Data and methodology

Aging macular degenerative disorder (AMD) is a progressive retinal condition that predominantly affects individuals over the age of 50, exerting a significant impact on eyesight. Visual impairments associated with AMD encompass distorted and blurred vision, manifesting as warped straight lines and the emergence of dark spots in the central field of vision, often leading to the alteration of color perception. As such, preemptive measures against ocular diseases hold paramount importance in sustaining clear and vibrant vision across our lifespan. In recent

years ([Avolio et al., 2023](#)), the integration of convolutional neural networks (CNNs) has showcased considerable promise in the realm of medical image processing. Notably, CNNs exhibit an exceptional capacity to autonomously recognize and extract pertinent features from images. To outline the proposed methodology for AMD eye disease detection, [Fig. 1](#) elucidates the sequential steps involved in the development of CNN-based approaches.

3.1. Data collection

The normal class signifies the absence of any specific eye disease or condition, representing optimal eye function and unimpaired vision. Diabetes, characterized by elevated blood sugar levels, can result in diverse ocular complications. Notably, diabetic retinopathy, a diabetic ocular disorder, arises when retinal blood vessels suffer damage due to high blood sugar levels ([Alyoubi et al., 2021](#)). Age-associated macular degeneration (AMD), primarily affecting older individuals, involves progressive macular deterioration, impacting central vision and causing blurred or distorted vision ([Acharya et al., 2016](#)). Cataracts, a common age-related condition, lead to clouding of the eye's crystalline lens, resulting in inconsistent or hazy vision ([Zaki et al., 2022](#)). [Fig. 2](#) displays distinct categories of fundus ophthalmology, encompassing Normal, Cataract, AMD, and Diabetes.

For the robust training of a CNN model, a diverse and well-annotated dataset containing both AMD and non-AMD eye images is imperative. The dataset employed in this study comprises a total of 2000 images, compiled by evaluating image quality from six public datasets, namely ODIR ([Larxel, 2020](#)), DR-200 ([Dugas et al., 2015](#)), Fundus Dataset ([Saeed & Rimsha, 2021](#)), RFMiD ([Awsaf, 2020](#)), ARIA ([Eyecharity.Com Is for Sale, 2023](#)), and Eye_Diseases_Classification ([Doddi, 2022](#)). The assessment of quality was conducted using contour techniques ([Watson, 2013](#)), focusing on edge sharpness and clarity, crucial aspects in human perception of image quality. These criteria encompassed illumination level, structural visibility, color contrast, and direct eye image. [Fig. 3](#)

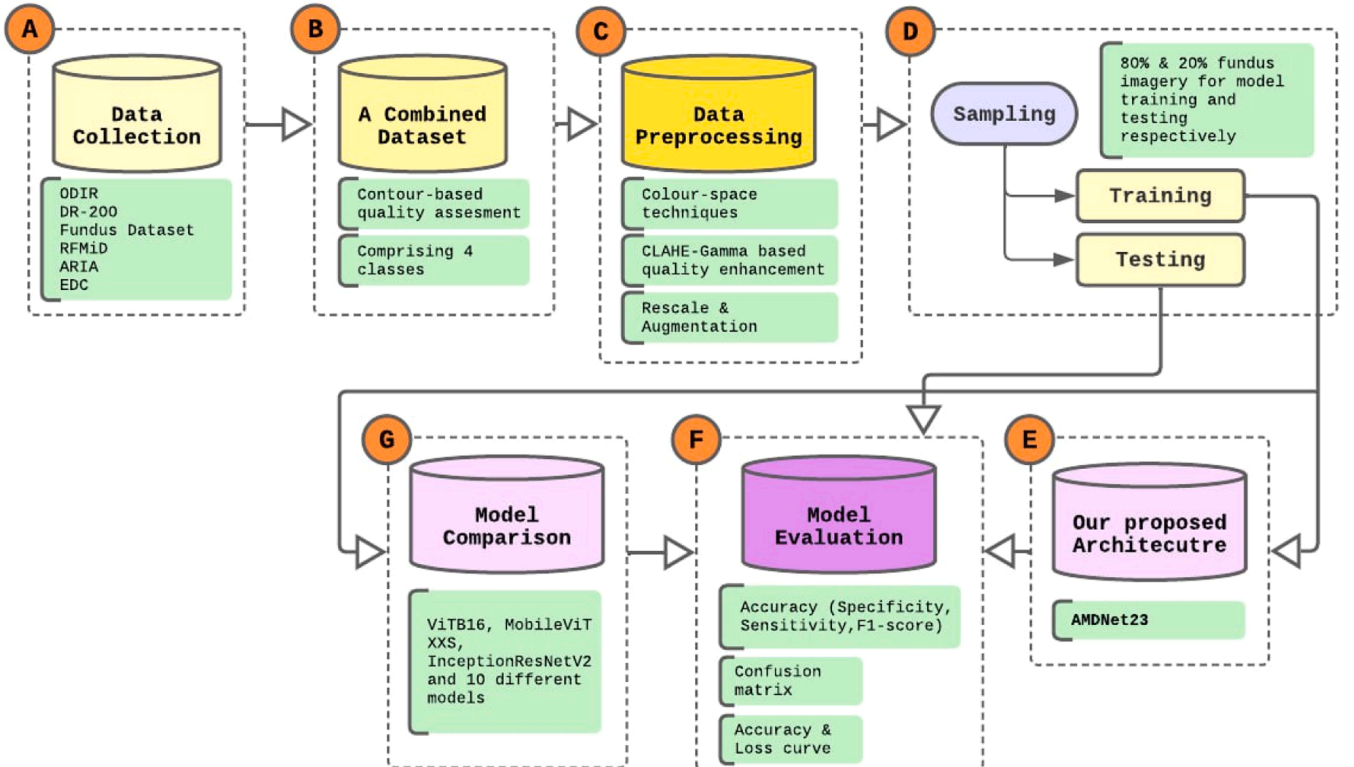


Fig. 1. Overall proposed methodology.

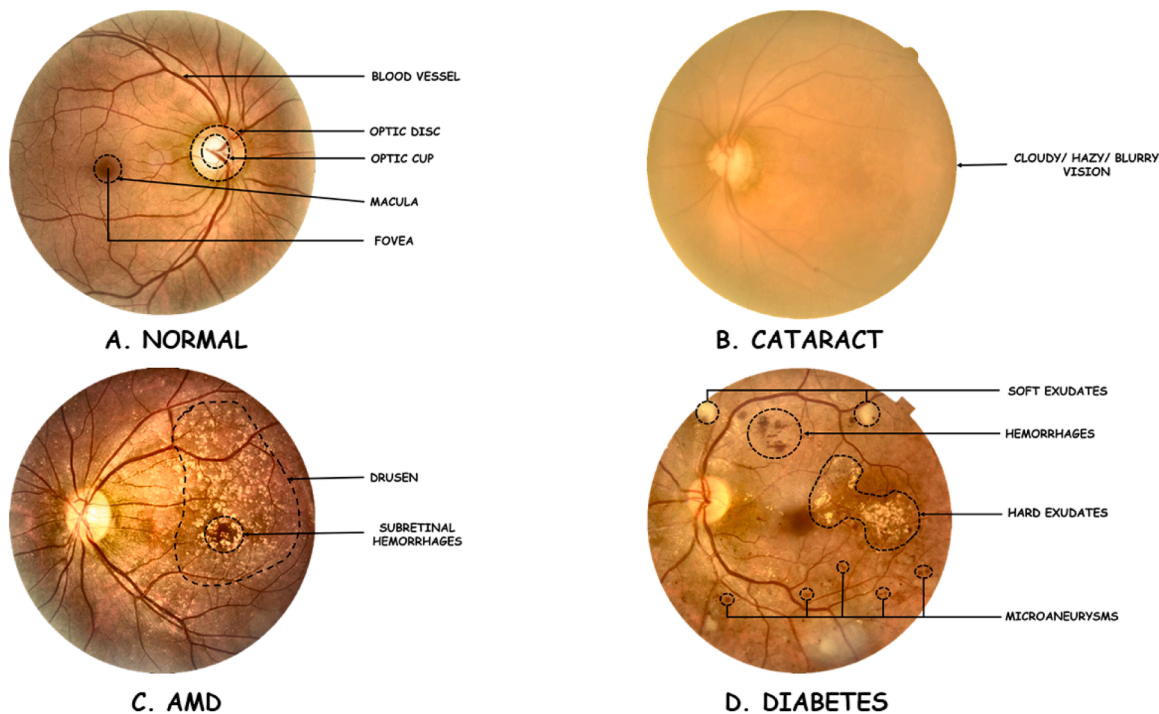


Fig. 2. Types of fundus ophthalmology.

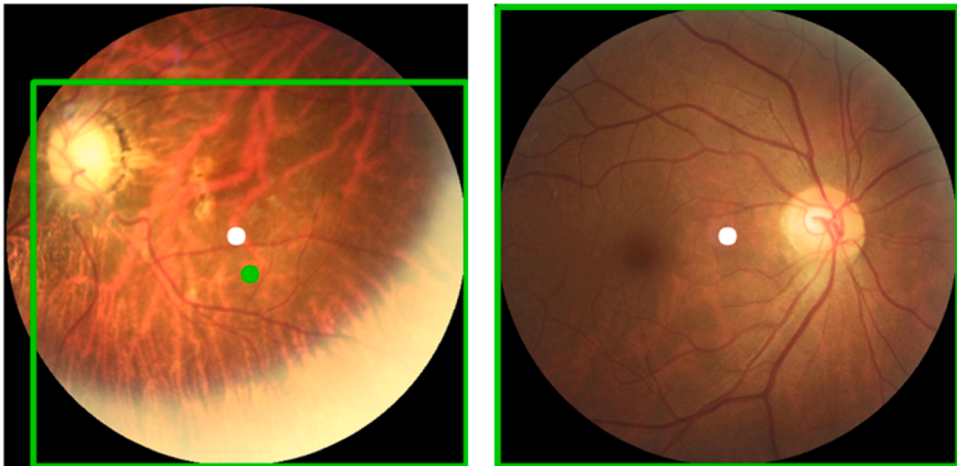


Fig. 3. Contour-based approach.

presents a Contour-based approach with a sample image demonstrating the evaluated image quality.

As observed, images with sharp, distinct edges correlate with high-quality visuals, while blurry or distorted edges signify lower quality. Such subpar images would detrimentally affect the machine’s interpretation. Our dataset encompasses four classes: Normal, Diabetes, AMD, and Cataract, each containing 500 images. The gathered dataset was partitioned into distinct sets to facilitate the training and evaluation of the proposed system. The training phase utilized 80 % of the dataset, while the remaining 20 % was employed for comprehensive evaluation and analysis. Table 1 outlines the available and chosen image quantities (in the first bracket) from the aforementioned six public datasets.

3.2. Data pre-processing

A distinctive highlight of our proposed approach lies in its pre-processing prowess, which focuses on enhancing image quality using

Table 1				
Summary of accessible and selected images from public datasets.				
Datasets	Normal	Diabetes	AMD	Cataract
ODIR	2873(168)	1608(300)	266(266)	293(256)
DR-200	1000(332)	1000(150)		
Fundus Dataset			46(46)	100(44)
RFMiD		376(50)	100(100)	
ARIA			101(88)	
Eye_Diseases_Classification				1038(200)

innovative techniques, setting it apart from conventional cataract disease detection methods. Diverging from earlier works, we introduced novel preprocessing methods in our study. The dataset underwent pre-processing in diverse color spaces (illustrated in Fig. 4) to extract features and elevate model effectiveness. Notably, among the RGB(G), HSV (V), and LAB(L) color spaces, the LAB(L) color space emerged as the

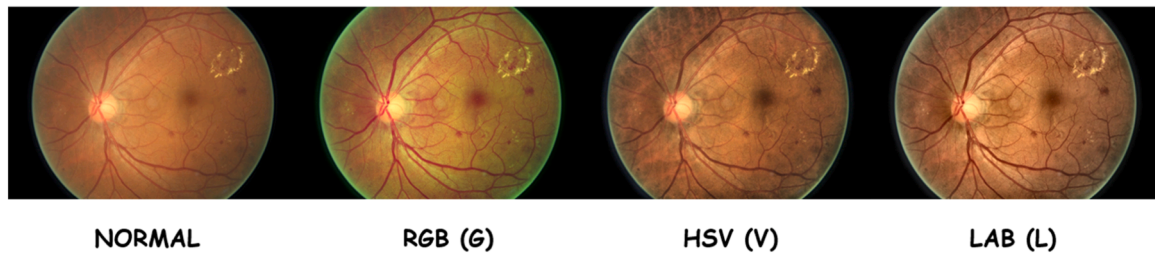


Fig. 4. Color spaces.

preferred choice due to its distinct visibility of vessels.

Subsequently, we employed preprocessing algorithms such as CLAHE and Gamma correction (Rahman et al., 2021) to enhance image quality by fine-tuning brightness and contrast. We conducted extensive parameter experimentation, eventually achieving satisfying outcomes with CLAHE (2.0, (8,8)). For gamma values of 0.5, the images tended to darken, while gamma values of 2.0 led to fading. To address this, CLAHE was employed to enhance regional contrast, rendering images more visually appealing and informative (Sahu et al., 2019). Fig. 5 visually presents the outcomes of our algorithmic experimentation, culminating in the selection of CLAHE (2.0, (8,8)) for our proposed model.

A histogram (Chugh et al., 2022) comparison between the implemented Context-limited adapted equalization of histograms (CLAHE) preprocessing algorithm and a regular image in Fig. 6 elucidates the impact of CLAHE in enhancing local contrast and improving image

quality.

Retinal fundus imaging utilizes diverse technologies for comprehensive eye assessment. Color fundus photography captures high-resolution images for general examination. Fluorescein angiography highlights blood vessels, aiding in vascular abnormality detection. Optical coherence tomography (OCT) creates cross-sectional images, useful for conditions like macular edema. Wide-field imaging provides a broader view, crucial for peripheral pathology. Autofluorescence imaging detects retinal dystrophies, while infrared imaging reveals structures unseen in traditional images, beneficial for retinal detachment cases. These technologies collectively enhance diagnostic capabilities in ophthalmology.

Through histogram comparison, one can discern the alterations in pixel intensity distribution pre and post the application of CLAHE. The histogram of the regular image showcases a relatively uniform pixel

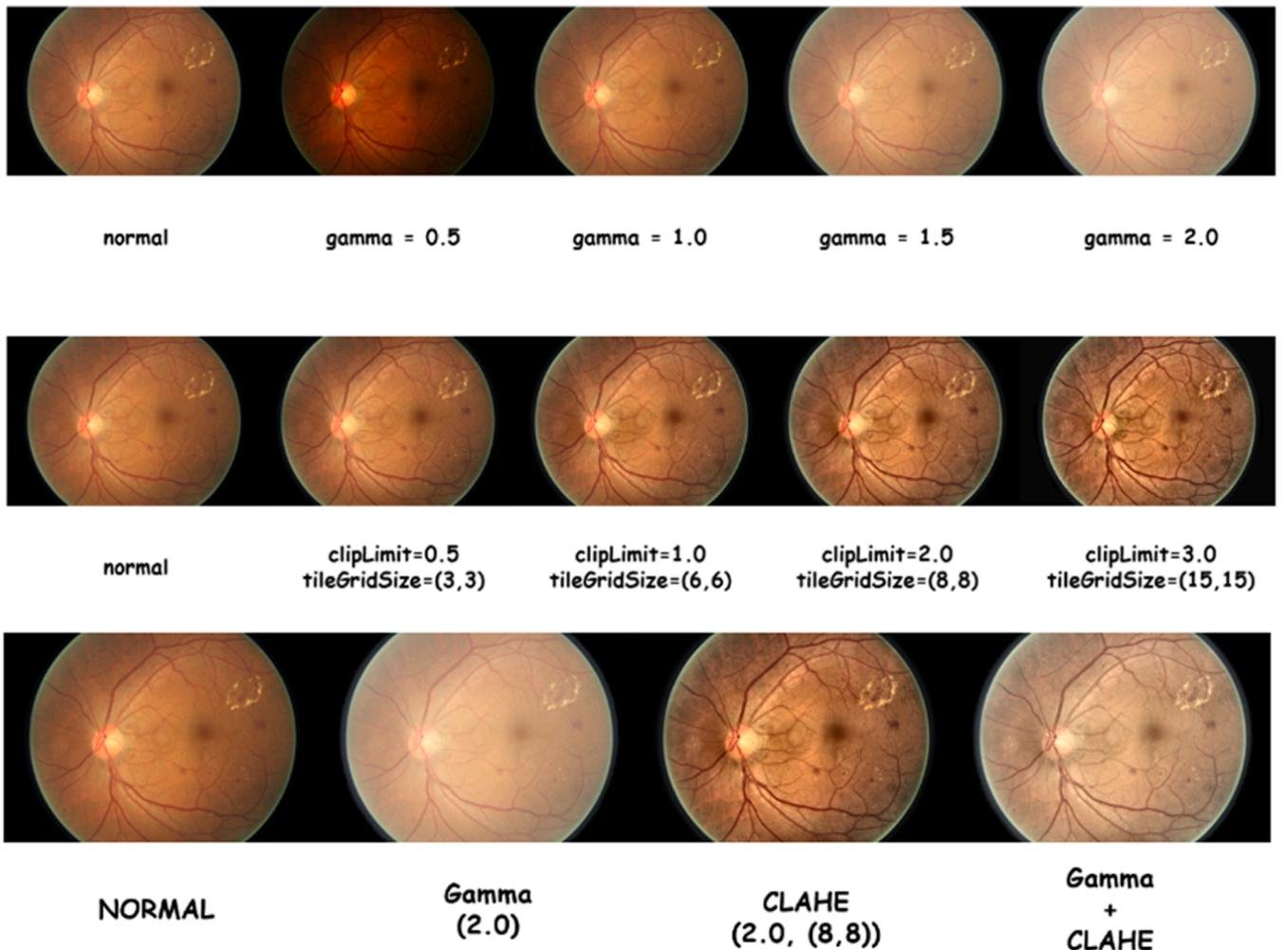


Fig. 5. Gamma and CLAHE based quality enhancement.

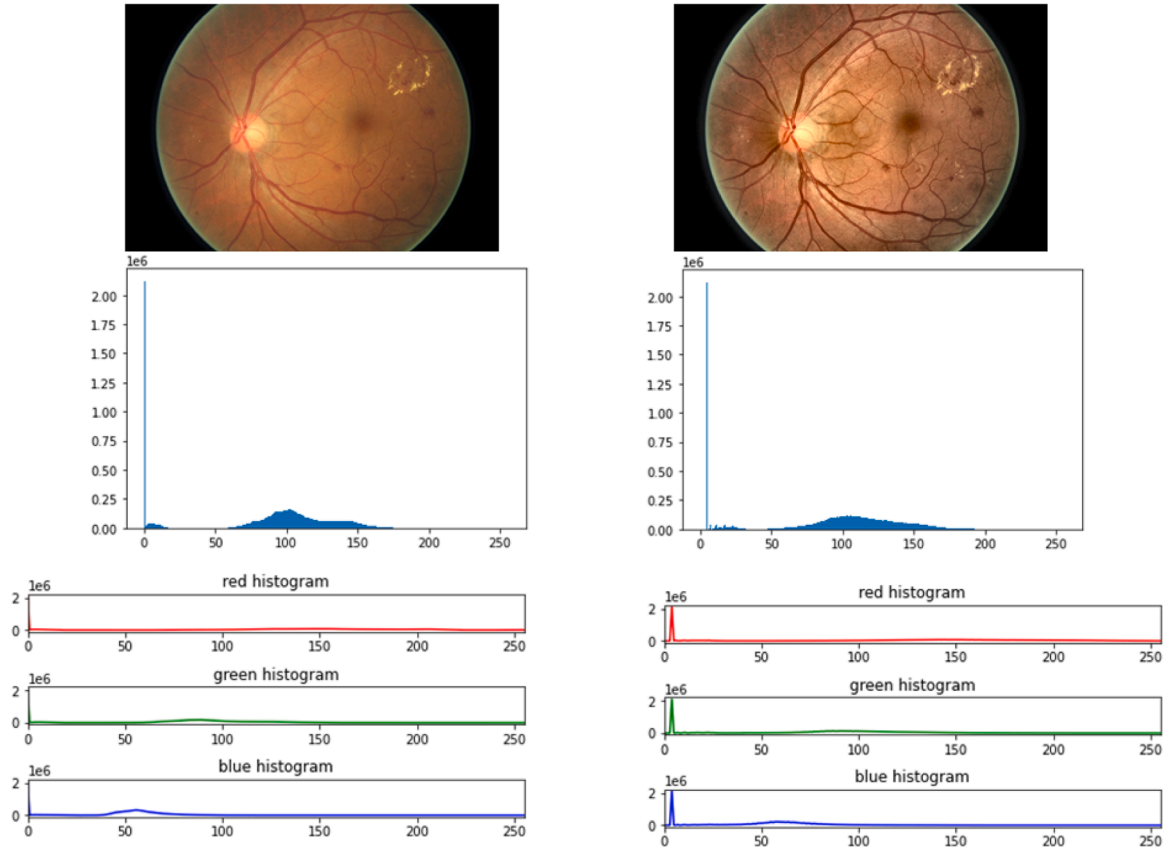


Fig. 6. Histogram comparisons.

intensity distribution, with variations influenced by image content. With CLAHE integrated into preprocessing, contrast enhancement is locally adapted, proving notably efficient in amplifying contrast within regions of diverse intensities. This process unveils concealed details and textures that may have been concealed in the initial image.

3.3. Statistical metrics

To gauge the efficacy of image preprocessing quality, Table 2 presents measurements of metrics including Mean Square Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM) (Sara et al., 2019). These metrics facilitate a comparison between the preprocessed image and the original image, evaluating the degree of distortion or resemblance.

Mean Squared Error (MSE): MSE generates the resultant mean squared disparity between the preprocessed and original image's pixel values. Lesser MSE reading indicate greater similarity between the images. MSE is calculated using the formula:

$$MSE = \frac{1}{m * n} \sum_{x=0}^{m-1} \sum_{y=0}^{n-1} [I(x, y) - K(x, y)]^2 \quad (1)$$

where $m * n$ represents the image dimensions, $I(x, y)$ and $K(x, y)$ indicate preprocessed image's pixel values and original images at coordinates (x, y) .

Table 2
Comparison metrics for preprocessed images.

Images	MSE	PSNR	SSIM
2376_left.jpg	2388.13	14.35	0.48
84_right.jpg	2189.98	14.72	0.65
980_right.jpg	927.54	18.45	0.53
71_left.jpg	709.37	19.62	0.54

$y)$.

Peak Signal-to-Noise Ratio (PSNR): The optical appealing of preprocessed photographs is frequently assessed utilizing the PSNR measure. It calculates a measure of the peak power ratio of the signal's strength to noise, which assessed in decibels (dB). Increased PSNR values indicated a greater similarity between the images. PSNR is calculated using the formula:

$$PSNR = 10 \log_{10} \frac{MAX^2}{MSE} \quad (2)$$

wherein MAX is the highest pixel value that is permitted to be used (for example, 255 in 8-bit photographs).

Structural Similarity Index (SSIM): SSIM evaluates the luminosity, contrary, and structural similarities between the preprocessed image and original image. The value of 1 denotes complete similarity, using SSIM readings varying from -1 to 1 . Higher SSIM values indicate better similarity between the images. SSIM is calculated using a combination of mean, variance, and covariance of the image patches.

$$SSIM = \frac{(2\mu_x\mu_y + c1)(2\sigma_{xy} + c2)}{(\mu_x^2 + \mu_y^2 + c1)(\sigma_x^2 + \sigma_y^2 + c2)} \quad (3)$$

where $c1$ and $c2$ are constants to prohibit division by zero, σ and μ stand representing the standard deviations and mean respectively.

By computing MSE, PSNR, and SSIM before and after image preprocessing, we can assess the effectiveness of the techniques in maintaining image quality and reducing noise, artifacts, or other undesirable effects. Lower MSE values, higher PSNR values, and greater SSIM scores denote improved preservation of image quality.

3.4. Transfer learning based approach

The underlying idea is to use a pre-trained model on a large dataset for a specific task and then fine-tune it on a smaller dataset for a related task. This process allows the model to capture generic features from the initial task, which can be beneficial for the new task. The shared knowledge reduces the need for extensive training on the new task, making the learning process more efficient and effective. In our study, we harnessed the potential of transfer learning (Yoo et al., 2019) by integrating a comprehensive set of 13 pre-trained models, each offering distinct technical attributes. These models, including ViTB16, MobileViT_XXS, InceptionResNetV2, EfficientNetB7, EfficientNetB6, DenseNet121, DenseNet169, DenseNet201, InceptionV3, MobileNetV2, VGG16, VGG19, and ResNet50, were employed to enhance the accuracy and robustness of our approach. By leveraging the knowledge extracted from their diverse architectures, we aimed to elevate the performance of our AMD disease detection system.

ViTB16 (Vision Transformer Base): A transformer-based architecture designed for image classification, utilizing self-attention mechanisms to capture relationships between image patches.

MobileViT_XXS (Mobile Vision Transformer Extra Small): A lightweight variant of Vision Transformer designed for mobile and resource-constrained devices, employing compact model size and efficient attention mechanisms.

EfficientNetB6 and EfficientNetB7: Both members of the EfficientNet family, these models embody the principles of compound scaling, meticulously balancing depth, width, and resolution for optimal performance. By efficiently utilizing resources, these architectures offer advanced capabilities for image classification tasks, catering to a wide range of applications.

DenseNet121, DenseNet169, DenseNet201: Dense Convolutional Networks featuring dense connections, allowing each layer to receive feature maps from all preceding layers, enhancing gradient flow, feature reuse, and parameter efficiency (Liu et al., 2019).

InceptionResNetV2: A hybrid architecture combining the multi-scale feature extraction of Inception modules and the residual connections of ResNet, resulting in improved accuracy and efficient training (Grassmann et al., 2018).

InceptionV3: A variant of Inception architecture incorporating factorization concepts and auxiliary classifiers to extract multi-scale features effectively (Chea & Nam, 2021).

MobileNetV2: A lightweight architecture tailored for mobile devices, leveraging depthwise separable convolutions and linear bottlenecks to achieve high efficiency (Domínguez et al., 2023).

VGG16, VGG19: VGG (Visual Geometry Group) models comprising 16 and 19 layers, characterized by a uniform architecture using small convolutional kernels and multiple max-pooling layers (Zang et al., 2023).

ResNet50: A Residual Network with 50 layers, introducing skip connections to address the vanishing gradient issue, enabling deeper architectures without performance degradation.

3.5. Proposed approach: hybrid CNN-LSTM model

The fusion of Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks constitutes a CNN-LSTM architecture, leveraging the image feature extraction capabilities of CNNs and the temporal modeling strengths of LSTM networks (Muni Mishra et al., 2022). Prior to inputting images into the model, a diverse range of augmentation techniques was applied to enrich the training dataset. These included randomized horizontal and vertical flipping, each with a 50 % probability, along with random brightness adjustments spanning from -0.1 to +0.1. Further diversity was introduced through random contrast adjustments ranging from 0.8 to 1.2, as well as random saturation adjustments between 0.8 and 1.2. Subtle color variations were achieved through random hue adjustments. Additionally,

translation-based width and height shifting was conducted within specific ranges to augment the dataset. This augmentation strategy enhances data diversity, thus improving the model's capacity to diagnose novel data instances (Lilhore et al., 2022). The model is constructed with a depth of 23 layers. Fig. 7 elucidates the architectural design of the innovative hybrid AMDNet23 model proposed for addressing AMD, while Table 3 details the corresponding configurations.

Input Layer: The CNN-LSTM model receives a dataset of eye images obtained from patients. These input images are structured as a tensor denoted as X , possessing dimensions (N, W, H, C) , where N signifies the count of eye images, and W and H denote the width and height of the images, respectively. In our case, the images were standardized to a size of 256×256 . C represents the count of color channels within each eye image. The tensor X is subsequently fed into the input layer of the model.

Convolutional Layers (conv2d): Following the initial input layer, the model incorporates a sequence of convolutional layers (Zang et al., 2023). Each convolutional layer employs a set of adaptable filters to analyze the input images. The outcome, referred to as the feature map from the i^{th} convolutional layer, is denoted as F_i , where i varies within the range of 1 to n . The computation of the output feature map F_i can be expressed as:

$$F_i = \text{Conv2D}(X, W_i) + b_i \quad (4)$$

In the above equation, Conv2D signifies the convolution operation, W_i denotes the learnable parameters (weights) specific to the i^{th} convolutional layer, and b_i represents the corresponding biases for that layer. The output feature maps F_i possess spatial dimensions (W', H') and C' channels.

The model was structured with six convolutional blocks. The initial four blocks encompassed 2 convolutional layers and 1 batch normalization layer each. The filter sizes for these blocks were 32, 64, 128, and 256, employing a kernel size of 3×3 . The fifth and sixth convolutional blocks included 3 convolutional layers and 1 batch normalization layer each, with all convolution layers within these blocks having 512 filters.

Pooling Layers: Following the convolutional layers, pooling layers are commonly employed for downsampling the feature maps (Muni Mishra et al., 2022). The feature maps obtained after pooling are denoted as P_i , with i ranging from 1 to p (total number of pooling layers). Each pooling layer conducts downsampling on the input feature maps. After the application of all pooling layers, the resultant feature maps are represented as P_p and possess spatial dimensions (W'', H'') and C'' channels. For the max-pooling layers within all convolutional blocks, a pool size of 2×2 was utilized.

LSTM Layer: The evolution and significance of long short-term memory (LSTM) networks are emphasized within the system. LSTM is a development over conventional Recurrent Neural Networks (RNNs) that aims to mitigate the vanishing gradient problem, which previously hindered the effective training of RNNs on lengthy sequences (Lilhore et al., 2022). LSTM introduces memory cells to retain information over extended periods, enabling it to capture long-term dependencies within input data. This mechanism empowers LSTMs to maintain a cell state that can retain or selectively discard information through input gates, forget gates, and output gates, forming the three key components. Mathematically, LSTM cell computations are expressed as follows, where 't' denotes the current time step, x_t signifies the input at time t , h_t represents the previous hidden state, and c_t indicates the cell state:

$$i_t = \sigma(W_i[x_t, h_{t-1}] + b_i) \quad (5)$$

$$\tilde{C}_t = \tanh(W_c[x_t, h_{t-1}] + b_c) \quad (6)$$

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t \quad (7)$$

The input gate (Eq. (5)) uses a sigmoid function to combine the previous output h_{t-1} and the present time input x_t , deciding the proportion of information to be incorporated into the cell state and Eq. (6)

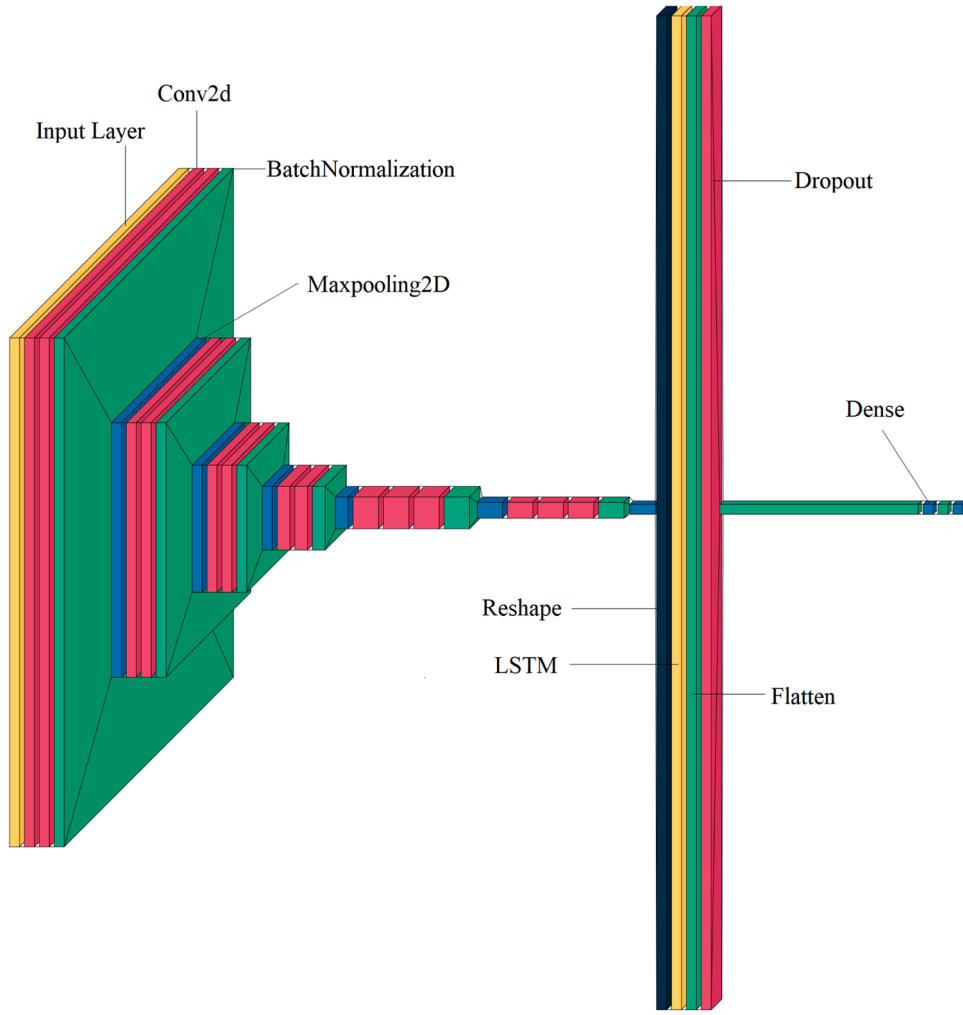


Fig. 7. Architecture of the proposed hybrid AMDNet23 model.

employs to obtain new information through the tanh layer to be added into current cell state $\tilde{C}_t$. The current cell state $\tilde{C}_t$, and long term information C_{t-1} are combination into C_t (Eq. (7)) whereas w_i determines the sigmoid output and $\tilde{C}_t$ determines to tanh output. “Forget” gate (Eq. (8)) investigates how much of the previous cell state should be retained and carried over to the next time step by assessing probability where W_f and b_f refers to the offset and weight matrix and offset respectively.

$$f_t = \sigma(W_f[x_t, h_{t-1}] + b_f) \quad (8)$$

The output gate of the LSTM investigates by h_{t-1} and x_t inputs following Eqs. (8) and (9) passed through the activation function to determine what portion of information to be appeared from the current LSTM unit at timestamp t .

$$o_t = \sigma(W_o[x_t, h_{t-1}] + b_o) \quad (9)$$

$$h_t = o_t \tanh(C_t) \quad (10)$$

In the above equation, W_o refers to the matrices of the output gate and b_o refers LSTM bias respectively.

Output Layer: The output layer provides the final prediction concerning the presence or absence of AMD in the input eye images. We can represent the input tensor for the output layer as ‘H_out’, obtained by reshaping ‘H_lstm’ to have dimensions (NT, D). An activation function, typically softmax, is applied after a dense layer in the output section. The dense layer takes ‘H_out’ as input and transforms it to produce the output tensor ‘Y’, which has dimensions (NT, K), where ‘K’ denotes the

number of output classes the model classifies.

3.6. Experimental setup

The setup encompasses the configuration of hyperparameters and hardware specifications. The training process was orchestrated with the following hyperparameters: The training data was divided into batches, each containing 32 samples. This batch size was chosen to balance memory efficiency and computational performance during training.

4. Results and discussions

In this section, we present the results and discussions stemming from our comprehensive evaluation of various models. Our assessment revolves around a detailed comparison of performance metrics, including Accuracy, Precision, Sensitivity, Specificity, and F1 score. A total of 14 distinct models were subjected to rigorous scrutiny, encompassing an array of established architectures. Notably, our study also introduces a novel Hybrid model designed to enhance performance.

The Table 4 showcases the evaluation metrics for various image classification models. At first glance, it’s evident that there is a significant disparity in performance across the models. Some models demonstrate impressive results, while others lag behind in various aspects. For accuracy, the top-performing models include AMDNet23 with 96.50 %, ViTB16 with 95.25 %, EfficientNetB7 with 92.75 %, and EfficientNetB6 with 92.75 %. These models have achieved high overall accuracy rates,

Table 3

Architecture configuration of the proposed hybrid AMDNet23 model.

Layer	Configuration
Input	Shape: (256, 256, 3)
1st Conv Block	Conv2D: Filters=32, Kernel=(3, 3), Activation='relu', Padding='same' Conv2D: Filters=32, Kernel=(3, 3), Activation='relu', Padding='same' BatchNormalization MaxPool2D: Pool_size=(2, 2)
2nd Conv Block	Conv2D: Filters=64, Kernel=(3, 3), Activation='relu', Padding='same' Conv2D: Filters=64, Kernel=(3, 3), Activation='relu', Padding='same' BatchNormalization MaxPool2D: Pool_size=(2, 2)
3rd Conv Block	Conv2D: Filters=128, Kernel=(3, 3), Activation='relu', Padding='same' Conv2D: Filters=128, Kernel=(3, 3), Activation='relu', Padding='same' BatchNormalization MaxPool2D: Pool_size=(2, 2)
4th Conv Block	Conv2D: Filters=256, Kernel=(3, 3), Activation='relu', Padding='same' Conv2D: Filters=256, Kernel=(3, 3), Activation='relu', Padding='same' BatchNormalization MaxPool2D: Pool_size=(2, 2)
5th Conv Block	Conv2D: Filters=512, Kernel=(3, 3), Activation='relu', Padding='same' Conv2D: Filters=512, Kernel=(3, 3), Activation='relu', Padding='same' Conv2D: Filters=512, Kernel=(3, 3), Activation='relu', Padding='same' BatchNormalization MaxPool2D: Pool_size=(2, 2)
6th Conv Block	Conv2D: Filters=512, Kernel=(3, 3), Activation='relu', Padding='same' Conv2D: Filters=512, Kernel=(3, 3), Activation='relu', Padding='same' Conv2D: Filters=512, Kernel=(3, 3), Activation='relu', Padding='same' BatchNormalization MaxPool2D: Pool_size=(2, 2)
LSTM Layer	Reshape: (16, 512) LSTM: Units=512, Activation='relu', Return_sequences=True, Trainable=False BatchNormalization Dropout: Rate=0.2
FC Layer	Flatten Dense: Units=64, Activation='relu' BatchNormalization
Output Layer	Dense: Units=4, Activation='softmax'
Compilation	Optimizer: Adam, Learning_rate=1e-3 Loss: Categorical_crossentropy Metrics: CategoricalAccuracy

indicating their proficiency in correctly classifying images across the dataset. AMDNet23 particularly stands out as it boasts the highest accuracy, showcasing its effectiveness in handling the classification task. However, a high accuracy score alone doesn't provide the complete picture. Looking at precision, which indicates the model's ability to minimize false positives, we notice that while AMDNet23 and ViTB16 maintain high precision levels (96.51 % and 95.26 % respectively), InceptionResNetV2 and MobileViT_XXS struggle with values of 36.26 % and 82.58 % respectively. This implies that the latter two models are more prone to labeling instances as positive when they are actually negative, impacting their reliability in applications where false positives are costly. Sensitivity, or recall, reveals a model's capability to detect positive instances accurately. Here, AMDNet23, ViTB16, EfficientNetB7, and EfficientNetB6 perform consistently well, all above 92.75 %. In contrast, InceptionResNetV2 trails significantly behind at 47.50 %, indicating its difficulty in capturing positive instances, potentially due to issues with feature extraction. Specificity, representing a model's aptitude in identifying negative instances, displays the models' varying performances. AMDNet23 remarkably achieves 99.32 %, showcasing its competence in discerning true negatives. On the other hand, InceptionResNetV2 and MobileViT_XXS exhibit relatively low specificity levels at 82.67 % and 98.28 % respectively, suggesting their tendency to incorrectly classify negative instances. The F1 score, which harmonizes precision and sensitivity, highlights the overall effectiveness of a model. AMDNet23, with its balanced precision and sensitivity, attains a high F1 score of 96.49 %. ViTB16 and EfficientNetB7 also maintain strong F1 scores, further establishing their robustness across multiple evaluation aspects.

Fig. 8 displays the confusion matrix for the AMDNet23 model, showcasing its performance in classifying a test set consisting of 400 images. The model achieved remarkable success by correctly classifying 386 out of the 400 images, indicating its high accuracy and proficiency in image classification tasks. In the matrix, the rows represent the true labels of the images, while the columns represent the labels predicted by the AMDNet23 model. Each cell in the matrix contains a count of instances corresponding to a specific true label and its predicted counterpart. This enables us to evaluate how well the model's predictions align with the actual classes.

Fig. 9 visually represents the journey of the AMDNet23 model in terms of accuracy and loss through its training process. The graph showcases the model's performance at its best, highlighting key insights into its learning and generalization capabilities. In the graph, the accuracy curve demonstrates how the model's accuracy evolves over successive epochs during training. As the epochs progress, the accuracy tends to increase, reflecting the model's improving ability to correctly classify images. The upward trajectory eventually stabilizes as the model converges, achieving its peak accuracy. This point is crucial as it signifies the model's optimal performance, showcasing its capacity to effectively capture underlying patterns in the data. The loss curve, on

Table 4

Performance comparison of different models.

Model	Accuracy	Precision	Sensitivity	Specificity	F1 Score
ViTB16	95.25 %	95.26 %	95.25 %	98.66 %	95.24 %
MobileViT_XXS	83.00 %	82.58 %	83.00 %	98.28 %	82.41 %
InceptionResNetV2	47.50 %	36.26 %	47.50 %	82.67 %	40.50 %
EfficientNetB7	92.75 %	92.94 %	92.75 %	98.97 %	92.62 %
EfficientNetB6	92.75 %	92.74 %	92.75 %	97.00 %	92.69 %
DenseNet121	82.25 %	82.41 %	82.25 %	95.00 %	81.46 %
DenseNet169	81.25 %	81.24 %	81.25 %	92.74 %	80.31 %
DenseNet201	84.75 %	84.45 %	84.75 %	93.46 %	84.52 %
InceptionV3	72.25 %	71.53 %	72.25 %	90.16 %	70.71 %
MobileNetV2	71.75 %	71.68 %	71.75 %	88.01 %	71.19 %
VGG16	89.75 %	89.82 %	89.75 %	94.68 %	89.78 %
VGG19	89.00 %	88.96 %	89.00 %	95.62 %	88.80 %
ResNet50	93.25 %	93.37 %	93.25 %	96.73 %	93.17 %
AMDNet23	96.50 %	96.51 %	96.50 %	99.32 %	96.49 %

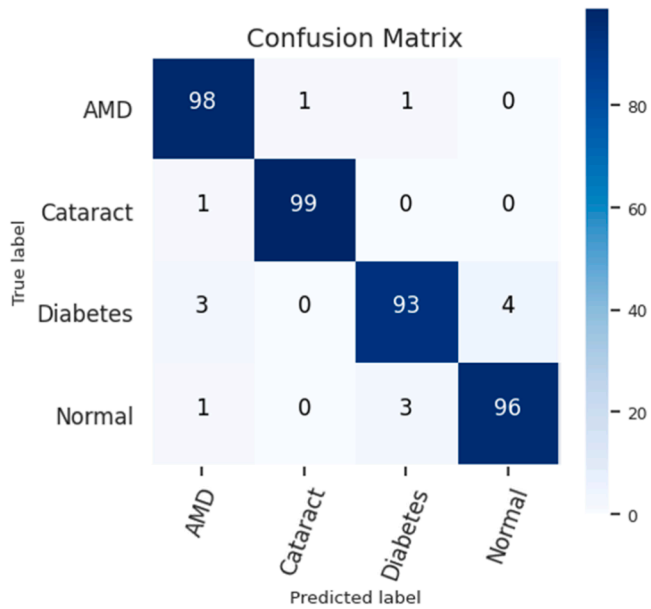


Fig. 8. AMDNet23 model's confusion matrix.

the other hand, illustrates the model's convergence towards minimal loss values. Loss, in this context, represents the discrepancy between the predicted labels and the actual labels in the training process. A decreasing loss curve signifies that the model is becoming better at minimizing prediction errors, aligning more closely with the ground truth labels.

The Table 5 presents a comprehensive comparison of recent advancements in image classification techniques. Authors have introduced diverse methodologies over the years, showcasing their models' capabilities in accurately categorizing images. Notable contributions include TK Yoo et al.'s VGG19-RF achieving an impressive 95 % accuracy with a 3-class classification. However, Huiying Liu et al.'s DeepAMD focuses on a more complex 6-class problem, achieving 70 % accuracy. Additionally, the proposed AMDNet23 method stands out in 2023 with a remarkable 96.5 % accuracy in a 4-class scenario. These insights highlight the continuous progress in the field, with each approach targeting varying complexities and achieving commendable results.

5. Study limitations and scope for future research

In examining the AMD detection model, it is essential to acknowledge certain limitations and consider avenues for future research. The model confronts challenges related to class imbalance, particularly

within minority classes, and awaits real-world clinical validation. Furthermore, its exclusive focus on age-related macular degeneration (AMD) restricts its versatility, prompting the potential inclusion of a broader spectrum of ocular diseases for a more comprehensive diagnostic framework. To improve comprehensibility, the recommendation is made for detailed explanatory documentation, covering preprocessing, network architecture, and parameter choices. A user-friendly interface is also suggested to enhance practical usability, particularly in clinical settings. Educational materials, including tutorials and case studies, are proposed to aid researchers and practitioners in navigating the nuances of the AMD detection framework. While our work contributes significantly to automated AMD diagnosis, there are acknowledged limitations. The reliance on a specific dataset may impact generalizability to diverse populations and imaging variations. Computational resource requirements and potential overfitting remain challenges. Future research could involve advanced preprocessing techniques, exploration of ensemble methods, and the incorporation of a more diverse dataset for improved real-world applicability. Collaborations with medical experts for clinical validation may pave the way for the model's adoption as a valuable tool in early AMD detection. These considerations collectively shape the study's limitations and offer directions for future research.

Table 5

State-of-the-art work comparison.

Author	Year	Method	No. of images	Accuracy
(Yoo et al., 2019)	2018	VGG19-RF	3000	3- Class 95 % accuracy
(Liu et al., 2019)	2019	DeepAMD	4725	6- Class 70 % accuracy
(Grassmann et al., 2018)	2020	Ensemble networks net	3654	13- class 63 % accuracy
(Chea & Nam, 2021)	2021	Optimal residual deep neural networks	2335	4- Class 85.79 % accuracy
(Domínguez et al., 2023)	2023	Transformer-based system	4896	3-Class 82.55 % accuracy
(Zang et al., 2023)	2023	Deep-Learning based aided system	Not specified	4-Class 80 % accuracy
Proposed Method	2023	AMDNet23	2000	4- Class 96.5 % accuracy

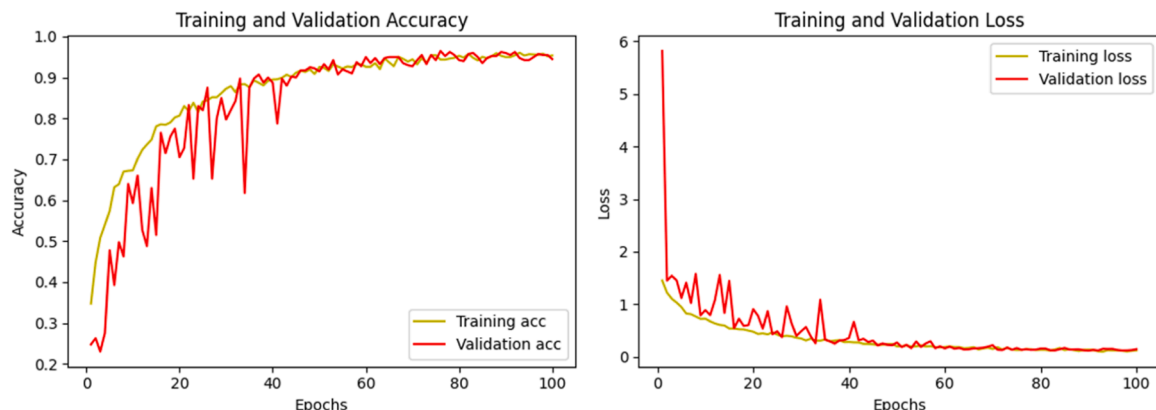


Fig. 9. Graphical illustration of best accuracy and loss curve of AMDNet23.

6. Conclusion

In this study, we introduced an advanced deep learning framework, AMDNet23, for automated Age-Related Macular Degeneration (AMD) detection in fundus images. Leveraging the integration of convolutional neural networks (CNNs) and long short-term memory (LSTM) mechanisms, our approach exhibited remarkable potential in enhancing the accuracy and efficiency of AMD diagnosis. By utilizing preprocessing techniques such as Adaptive Contrast Enhancement Algorithm (CLAHE) and Gamma correction, we improved the quality of fundus images, laying a strong foundation for accurate disease detection. The results of our experiments were promising. AMDNet23 demonstrated an impressive accuracy of 96.50 % in detecting AMD across a diverse dataset of 2000 fundus images, effectively outperforming 13 other pretrained CNN models. This not only highlights the efficacy of our proposed approach but also its significance in the realm of ocular health diagnostics.

CRedit authorship contribution statement

Md. Aiyub Ali: Conceptualization, Methodology, Investigation. **Md. Shakhawat Hossain:** Data curation, Formal analysis, Writing – original draft. **Md. Kawsar Hossain:** Software, Validation, Writing – original draft. **Subhadra Soumi Sikder:** Writing – original draft, Resources, Visualization. **Sharun Akter Khushbu:** Supervision. **Mirajul Islam:** Supervision, Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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